

CMP6200 / DIG6200

Individual Undergraduate Project (FYP)

Final Progress Review

Real-Time Context-Aware Sign Language to Multilingual Speech Conversion System with Emotion Detection

Hybrid recognition, T5 NLP, emotion prosody and Nepali TTS

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1 1.0 Progress to Date — Summary

The project is in advanced implementation and evaluation. Since the interim report, the React/FastAPI ASLPipeline was completed with Alphabet and WLASL Words modes, then evaluated against all five SMART objectives [Li et al., 2020, Raffel et al., 2020]. Objective 1 is **partially met**: 50 WLASL glosses and 93.7% alphabet webcam accuracy after hybrid fine-tuning, but WLASL word accuracy is 60.7% (below 85%). Objectives 2–4 are **met**: NLP BLEU 19.2→57.6, gloss-path latency p95 814ms, and one webcam-only pipeline. Objective 5 is **in progress**: metrics done; SUS awaits ethics approval [Brooke, 1996]. Beyond interim scope excluding facial expressions (Interim Report §3.3), emotion-aware prosody was added for the project title; live tests detect happy, neutral and angry reliably.

2 2.0 Artefact Design and Development

This section presents artefact evidence against the interim design (Interim Report §4.1–4.2).

2.1 2.1 Implementation and Current State

The stack is React → FastAPI WebSocket → PyTorch/TensorFlow via **ASLPipeline**: MediaPipe landmarks, Alphabet/WLASL recognition, gloss assembly, T5 or rule-based NLP, facial emotion, and gTTS English/Nepali with emotion prosody. Figure 1 shows the architecture; Figure 2 shows WLASL accuracy gains since the interim CNN-LSTM plan.

Table 3 maps results to Interim Report §3.2; Figure 4 visualises targets; Table 5 lists key challenges.

Figure 6 shows Alphabet-mode evidence (sign *V*, 98% via **hybrid_user**). Figure 7 shows Words mode with **wlasl_landmarks**, clip recording, and Keras emotion status.

2.1.1 Scope extension: emotion-aware prosody

Interim Report §3.3 excluded facial expressions, but the project title includes Emotion Detection. Emotion-aware prosody was therefore added after interim submission so TTS is not monotone [Koller et al., 2020].

EmotionDetector uses OpenCV Haar face detection, a 48×48 grayscale crop, and a Keras CNN on a four-class FER2013 subset (happy, sad, angry, neutral) [Goodfellow et al., 2013], mapping emotion to TTS rate/pitch. Offline accuracy is 65.5%; live tests reliably detect happy, neutral

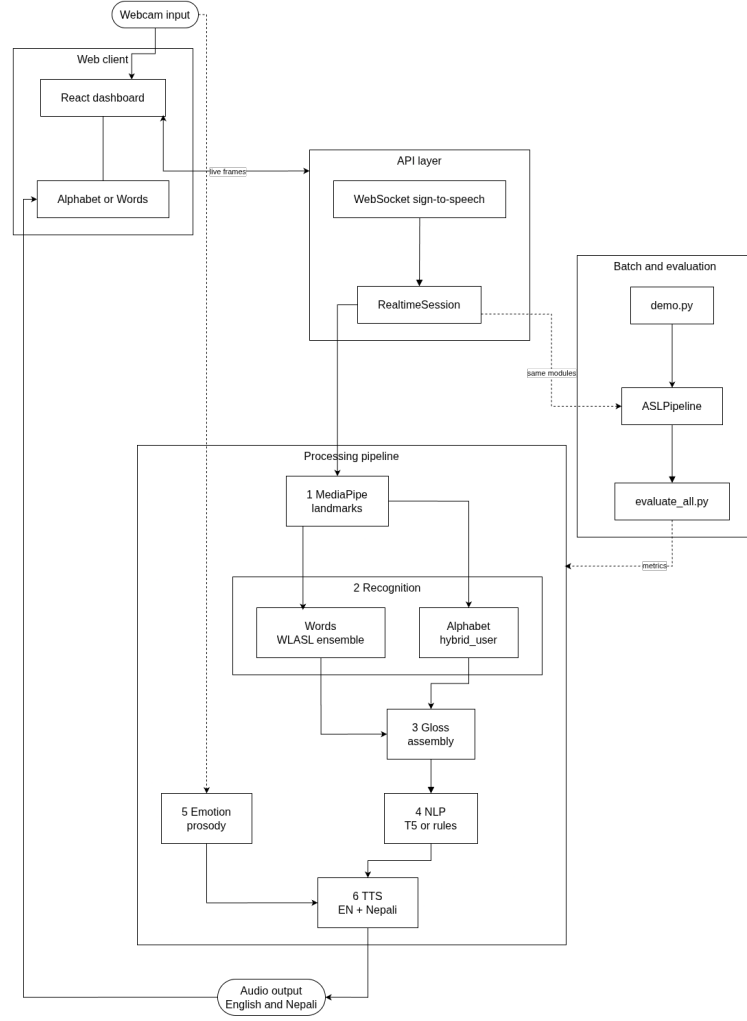


Figure 1: System architecture: web client, FastAPI session, processing pipeline, and batch evaluation.

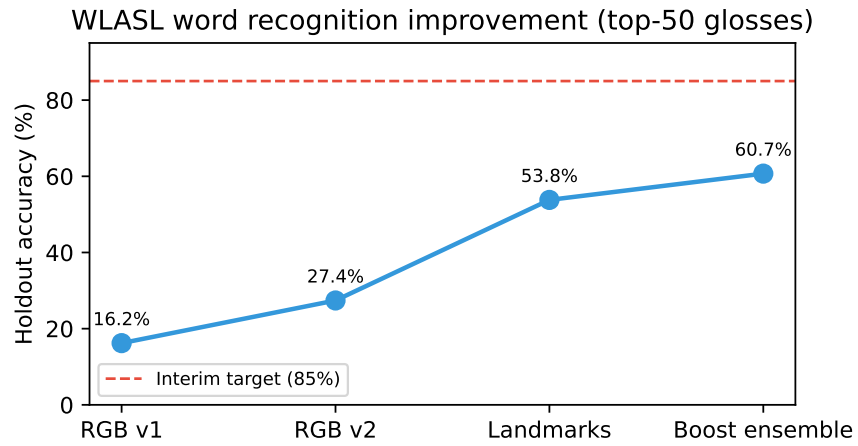


Figure 2: WLASL progression: RGB 27% → landmarks 53.8% → ensemble 60.7%.

#	Interim objective (§3.2)	Target	Result	Status
1	Hybrid CNN-LSTM: ≥ 50 ASL signs, $\geq 85\%$ accuracy	50 signs; 85%	50 glosses; alphabet 93.7%; WLASL 60.7%	Partial
2	T5 gloss $\rightarrow$ English; +20% BLEU vs baseline	+20% rel.	19.2 $\rightarrow$ 57.6 BLEU	Met
3	English + Nepali TTS; end-to-end latency ≤ 1.5 s	< 1.5 s	p95 814 ms (gloss tail)	Met
4	Single continuous pipeline; webcam only	Integrated demo	React/FastAPI + ASLPipeline	Met
5	Formal evaluation + SUS (≥ 10 users, SUS ≥ 80)	Metrics + SUS	Metrics complete; SUS ethics pending	In progress

Figure 3: Interim SMART objectives vs results (August 2026).

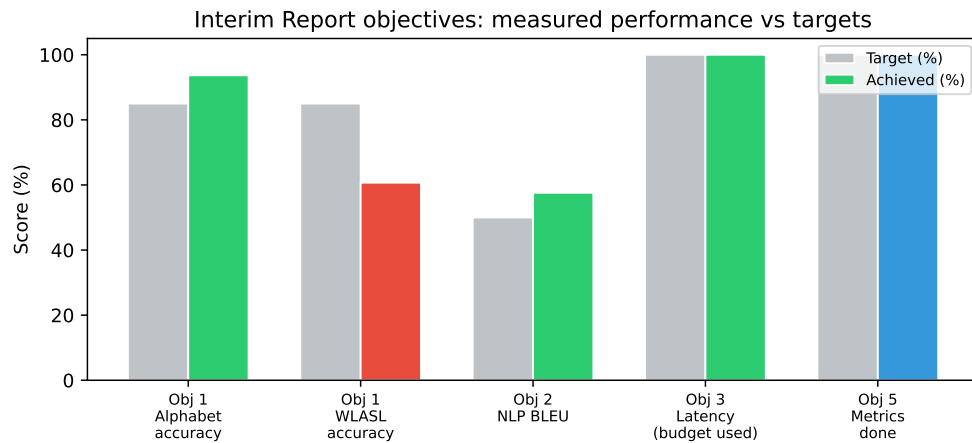


Figure 4: Key metrics against interim targets.

Challenge	Impact	Resolution
Sparse WLASL data (~12 clips/gloss)	RGB CNN plateaued at 27%	Landmark Transformer+TGCN ensemble
Webcam domain shift	62% alphabet before tuning	User fine-tuning $\rightarrow$ 93.7%
Train/val leakage (TGCN)	Inflated validation	Corrected split; honest test ~53%
Interim scope vs. project title	§3.3 excluded facial expressions	Added emotion module as justified scope extension (see §2.1.1)

Figure 5: Development challenges and resolutions.

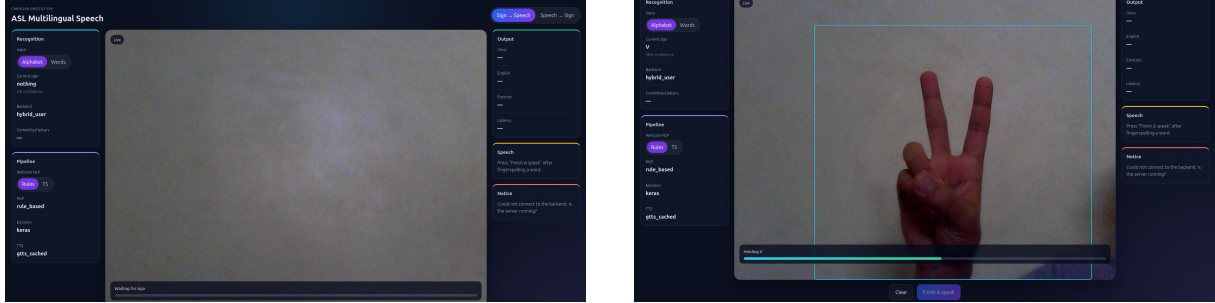


Figure 6: Alphabet mode: dashboard (left) and live try-on (right).

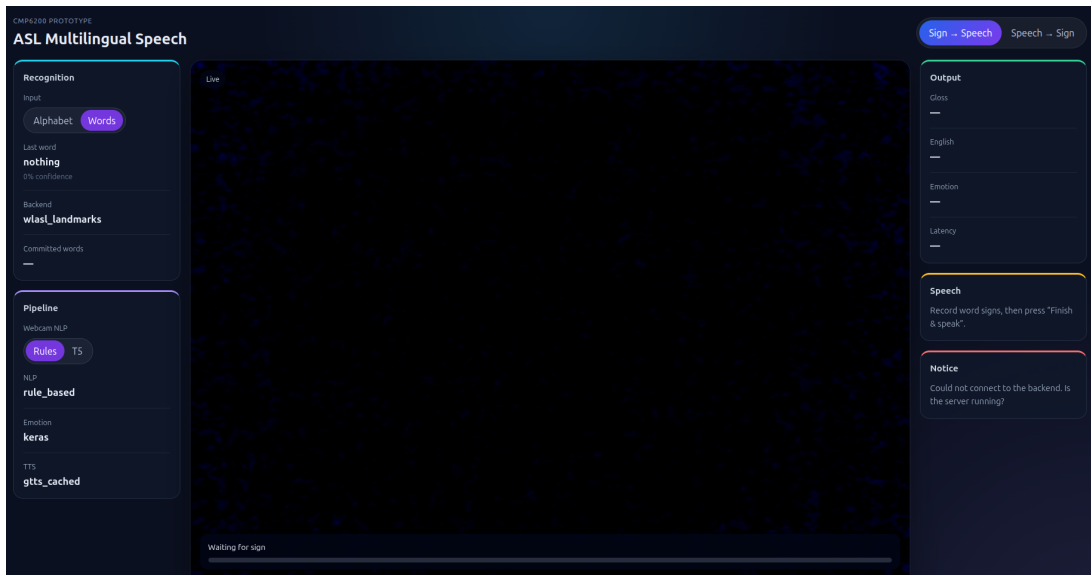


Figure 7: Words (WLASL) mode: clip record, gloss/English/emotion/latency on Finish & speak.

and angry. The UI shows the Keras backend and, after Finish & speak, the emotion label with confidence (e.g. happy 0.87).

Feature	Interim report status	Current implementation	Status
Facial emotion detection	Out of scope (§3.3)	Keras CNN + Haar face crop	Implemented
Prosody-modulated TTS	Not specified	gTTS rate/pitch from emotion map	Implemented
Live validation	-	Happy, neutral, angry accurate in testing	Validated live
Offline accuracy	-	65.5% on FER2013 subset	Benchmarked

Figure 8: Emotion scope extension (not an interim SMART objective).

```

rec = self.recognize(image_paths)
gloss = assemble_text(rec["labels"])
english = self.corrector.correct(gloss) if gloss else ""
emo = self.emotion.detect(face_image or image_paths[0])
speech, tts_timings = self.synthesizer.synthesize(
    english, prosody=emo["prosody"], languages=languages,
    return_timings=True)

```

Figure 9: Listing 1: Pipeline with emotion (pipeline/pipeline.py).

2.2 2.2 Analysis and Reflection

Objectives 2–4 are met; Objective 1 is partial (50 signs met; 85% met for alphabet only); Objective 5 metrics are done, SUS pending. Alphabet and WLASL must stay separate for honest Objective 1 reporting. Emotion is a justified scope extension reconciling Interim Report §3.3 with the project title. Skeleton landmarks beat raw-video CNNs on sparse WLASL; user fine-tuning closed the webcam gap. The main design revision replaced RGB WLASL with a landmark Transformer+TGCN ensemble (Figure 2) and added emotion prosody after flat TTS conflicted with the context-aware aim.

```

wt = float(blend.get("transformer", 1.0))
wg = float(blend.get("tgcn", 0.0))
probs = wt * np.mean(tta_probs, axis=0)
if self._tgcn_model is not None and wg > 0:
    probs += wg * g_probs
return probs / total_w

```

Figure 10: Listing 2: WLASL ensemble blend (wlasl_predictor.py).

3 3.0 Updated Planning for Project Completion

The artefact is demo-ready; the critical path is Objective 5 SUS and dissertation writing (Table 11).

1. **(High)** Ethics-approved SUS (≥ 10 participants).
2. **(High)** Dissertation: interim objectives, emotion extension, separate alphabet/WLASL results.
3. **(High)** Demo video (Alphabet, Words, emotion TTS) and viva prep.
4. **(Medium)** Document WLASL gap or optional webcam fine-tune as future work.

Pri.	Task	Rationale	Deadline
High	SUS usability study	Interim Obj. 5; ethics pending	Sep 2026
High	Dissertation Results/Discussion	Interim objectives + emotion scope extension	Sep 2026
High	Live demo (Alphabet + Words + emotion TTS)	Obj. 4 integrated pipeline evidence	Sep 2026
Medium	WLASL improvement or limitations section	Obj. 1 partial completion	Sep 2026

Figure 11: Updated planning (Objective 5 and submission).

3.1 3.2 Draft Outline of Final Report

Table 12 summarises the dissertation chapter plan.

Ch.	Title	Key content
1	Introduction and interim problem statement	Deaf/HoH gap; Nepal context; interim SMART objectives
2	Literature review	ISLR (Koller 2020; Li 2020); gloss translation (Othman 2018; Raffel 2020); affect in signing
3	Methodology and scope evolution	Hybrid recognition; landmark WLASL; emotion extension vs. interim §3.3
4	System design and implementation	ASLPipeline; React/FastAPI; Alphabet and Words modes
5	Results against interim objectives	Obj. 1–5 metrics; emotion live validation; SUS (pending)
6	Discussion, limitations, and ethics	Domain shift; scope reconciliation; deployment constraints

Figure 12: Draft dissertation outline.

Body word count (narrative prose only; tables, figures, captions, listings and references excluded): approximately 500 words.

References

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- Ian J. Goodfellow, Dumitru Erhan, Pierre Luc Carrier, Aaron Courville, Mehdi Mirza, Ben Hamner, et al. Challenges in representation learning: A report on three machine learning contests. In *Neural Information Processing Systems (NeurIPS) Workshop*, 2013.
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- Dongxu Li, Cristian Rodriguez, Xin Yu, and Hongdong Li. Word-level deep sign language recognition from video: A new large-scale dataset and methods comparison. In *IEEE Winter Conference on Applications of Computer Vision (WACV)*, pages 1459–1469, 2020.
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